ELASTICITY

(MANKIW CHAPTER 5)

C211: Global Economics for Managers

Western Governors University

ELASTICITY

- Measures the responsiveness of demand to changes in:

  - Changes in the price of the good

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- Changes in the income of consumers

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

  - Changes in the prices of a related good

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good 1}}{\text{Percentage change in the price of good 2}}$$

PRICE ELASTICITY

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- **Elastic** goods has a **Price Elasticity of Demand greater than 1**
  - These are goods for which, when the price changes, we change our quantity demanded a lot
  - See Ch5-2d

More likely to be elastic if:

- Goods with lots of substitutes
- Luxuries
- Longer Time Horizon

PRICE ELASTICITY

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- **Inelastic** goods has a **Price Elasticity of Demand less than 1**
  - These are goods for which, when the price changes, we change our quantity demanded by very little
  - See Ch5-2d

No (or few) Substitutes

Necessities

Shorter Time Horizon

NORMAL VS INFERIOR GOODS

- Normal Goods
  - Things we buy more of when our incomes increase

- Inferior Goods
  - Things we buy less of when our incomes increase

INCOME ELASTICITY

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

- Measures the responsiveness of Demand to changes in consumer income.

- But consumer demand will respond differently depending on if the good is a:

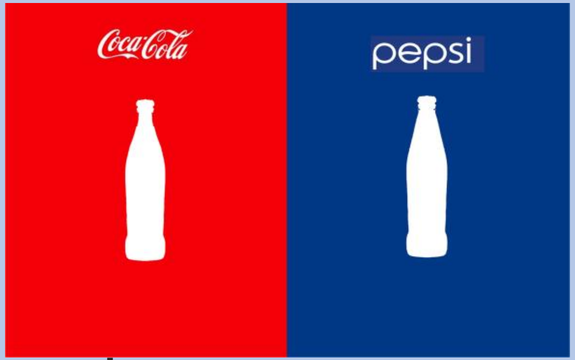
- **Normal Goods**
  - Buy more when our incomes increase
  - Positive Income Elasticity of Demand

- **Inferior Goods**
  - Things we buy less of when our incomes increase
  - Negative Income Elasticity of Demand

RELATED GOODS

- **Substitutes**—Two goods which do the same job or are similar in another way so they can be substituted for each other in consumption
  - Ex. T-shirts and Polo shirts; Apples and Oranges; Water and Soda





- **Complements**—Two goods which must be used together
  - Ex. Chips and Salsa; Coffee and Milk; Peanut Butter & Jelly

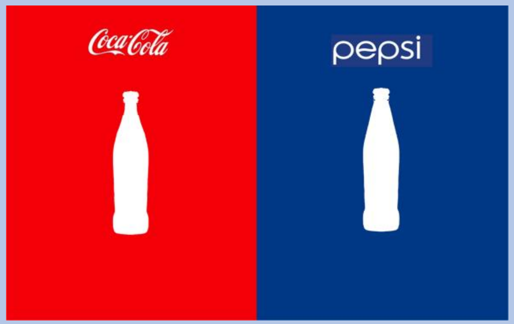


CROSS-PRICE ELASTICITY

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good 1}}{\text{Percentage change in the price of good 2}}$$

- Cross-Price Elasticity measures the responsiveness of demand to changes in the price of a related good.

- Substitutes
  - Buy more of the substitute when price increases
  - Positive Cross-Price Elasticity of Demand



- Complements
  - If the price of one goes up, we buy less of both
  - Negative Cross-Price Elasticity of Demand

SUPPLY

- Elasticity can work the same way for Supply

- Price Elasticity of Supply for example measures how much Quantity Supplied responds to a change in Price

- For example, housing supply is inelastic, so even when price changes the quantity supplied stays the same (in the near term).

- Or if it is the Supply of potato chips, it would be relatively easy to increase Supply in response to a price increase.